

Filling the static system

General

Static Systems

The magnet design does not include a permanently fitted syphon. A short magnet service syphon has to be fitted when filling with liquid helium. The service syphon is tooling only, and must be removed immediately after filling has been completed.

This procedure can be performed on magnets at field and magnets at zero field.

NOTE Only CSEs trained by Siemens in filling the magnets are allowed to perform this work.

NOTE It is recommended that fills are performed during service visits and before ramping up.
Do not exceed 95%.

NOTE If the magnet temperatures rise above the He quoted in the installation data, then a bottom fill may be required. Refer to ([Bottom filling the magnet](#)) Contact Magnet HSC for advice.

Venting system

The document describes the procedure for magnets with **Mk1** ([Magnet with Mk1 Venting System](#)) and **Mk2** ([Magnet with Mk2 Venting System](#)) venting systems.

Fig. 47 Mk1 OR105 Venting System

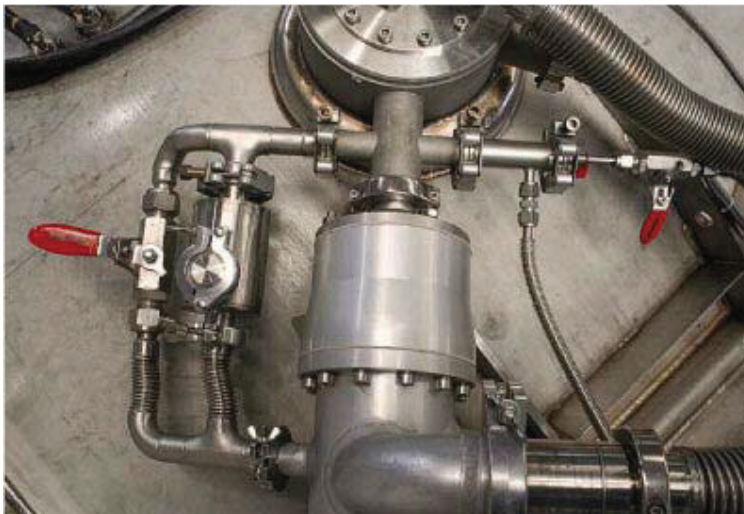
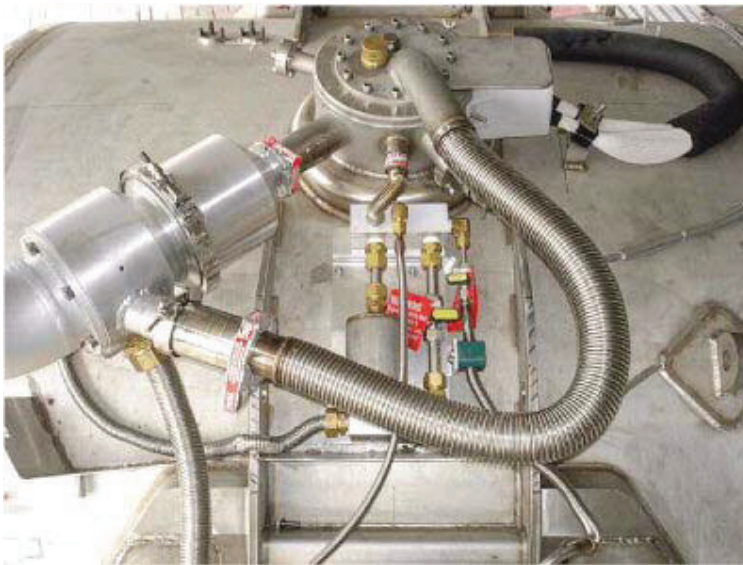


Fig. 48 Mk2 Venting System



Safety

CAUTION

Cryogenic liquids.

Failure to observe the following safety warning may result in cold burns.

- ➔ Protective clothing, gloves and eye protection must be worn. Filling should be performed by trained personnel only.

WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- ➔ Only non-magnetic equipment may be taken into the magnet room.
Never transport the GHe steel cylinder into the magnet room.

WARNING

Risk of asphyxiation.

Failure to observe the following safety warning may result in dizziness and loss of consciousness in personnel working with cryogen.

- ➔ Do not vent helium gas into the magnet room during helium transfer. As an exception, helium gas may be vented to cool the syphon.
Ensure that the turret venting kit is connected to quench pipe.
Oxygen monitors must be activated at all times.

CAUTION

Pressurized helium vessel.

Cold helium gas.

Failure to observe the following safety information may result in physical injury.

- ➔ Do not disconnect any of the fittings without first depressurizing the helium vessel.

NOTICE

Mishandling of the dewar.

- ➔ The dewar-relevant safety regulations of the manufacturer must be observed.

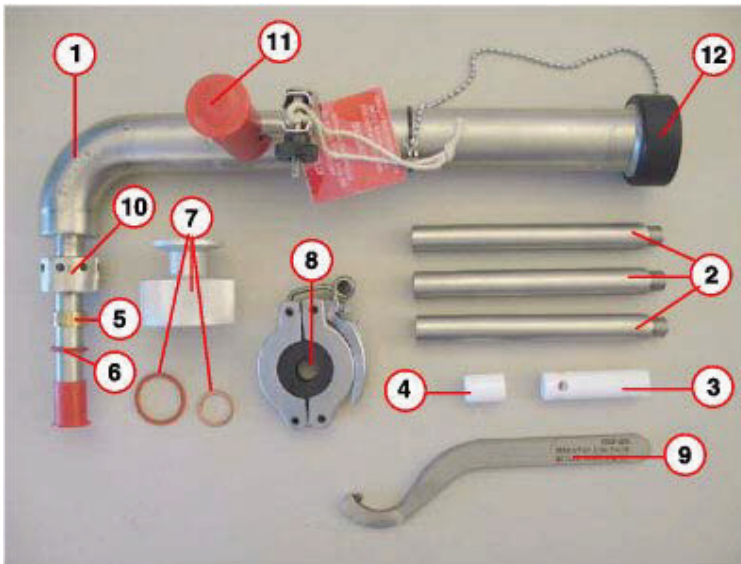
NOTE

Monitor the filling procedure continuously by checking transfer pressure and the fill level.
Identify hazardous areas with warning signs.

Equipment required

- Dewar syphon
- Dewar neck adaptor, clamps, seals, and hose
- Pressure service tool (part no **03861221**)
- Magnet service syphon kit (part no **08396363**)

Fig. 49 Magnet service syphon kit



- | | |
|---------|---------------------------------|
| Pos. 1 | Service syphon assy |
| Pos. 2 | 130mm syphon extension legs |
| Pos. 3 | 40mm PTFE syphon defuser |
| Pos. 4 | PTFE seal (bottom fill) |
| Pos. 5 | Brass spacer |
| Pos. 6 | Silicon 'O'-ring |
| Pos. 7 | LN2 To LHe adapter with O-rings |
| Pos. 8 | Syphon leg support clamp |
| Pos. 9 | Syphon 'C' spanner |
| Pos. 10 | Ring nut |
| Pos. 11 | Vacuum valve |
| Pos. 12 | Syphon end cap |

- Fig. 50 MSUP Display unit

- MSUP hand held display unit



- Heat gun
- Helium gas supply (at least 99.9% pure)
- Pressure regulator 200 bar/ 0 - 1.5 bar
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes
- Replacement copper seal for syphon port on the turret
- Service platform
- Either a room or personal O2 Monitor
- Gas bottle wrench

Mk2 venting only

- Syphon port adaptor (part no **10113551**)
- 30 mm non-magnetic syphon spanner

System preparation

Certain operations must be performed without delay to ensure efficient transfer:-

- Ensure that there is open and clear access to the magnet for helium dewar(s).
- Ensure that there is open and clear access to the syphon.
- Position the **Service platform** next to the magnet. The platform must be secured to the magnet frame using platform straps, or by using strong tie wraps.
- Fit the **hand held remote unit** to the MSUP to enable continuous monitoring of the pressure and LHe levels during this procedure. Ensure the MSUP is clearly visible.

Dewar preparation

This procedure describes the use of **top fill** helium dewars used in conjunction with the Siemens Magnet Technology dewar syphon.

It covers all dewars that are externally pressurized with helium gas.

NOTE

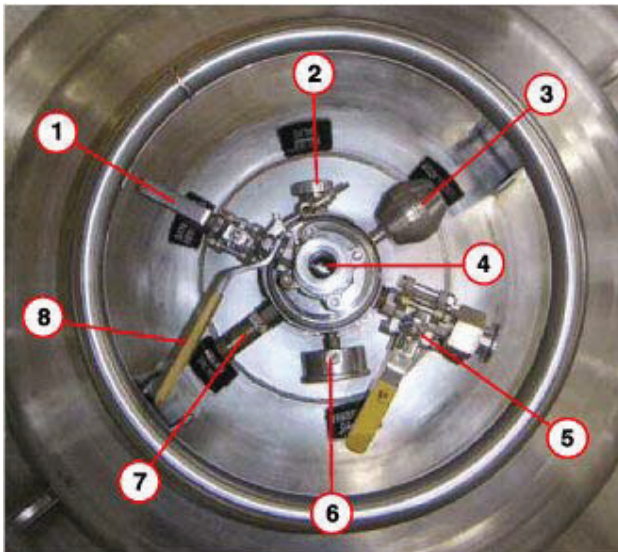
Refer to the manufacturers documentation when using side fill and/or self-pressurizing dewars.

NOTE

Read the dewar manufacturers safely guidelines and check the dewar valves before commencing the LHe fill.

- If the dewar is highly pressurized, release the pressure by slowly opening the **primary ball valve** ([5/Fig. 51](#)) to minimize the flash losses of helium.
This must be done in a well ventilated area - **NOT in the imaging suite.**
- Fit the **dewar syphon adaptor** ([3/Fig. 53](#)) to the top of the dewar.

Fig. 51 Dewar Ports (typical)



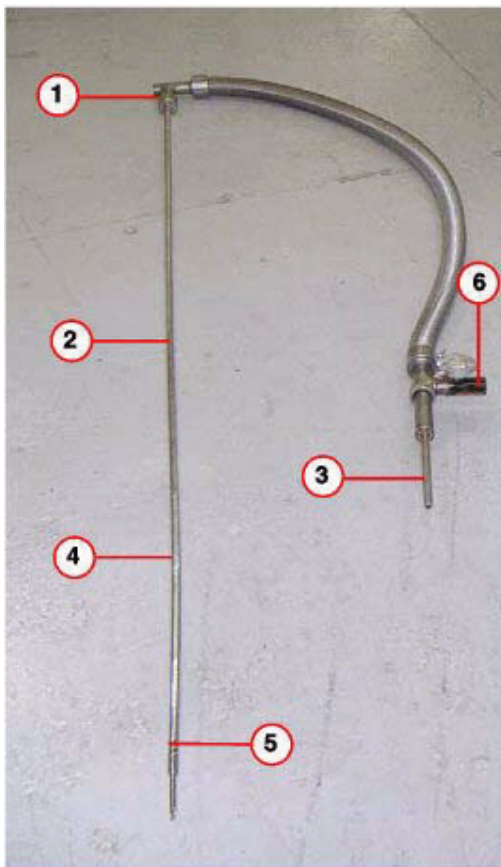
- Pos. 1 Low pressure relief valve (shown open)
- Pos. 2 Relief valve
- Pos. 3 Anti-oscillator
- Pos. 4 Syphon entry port (shown closed)
- Pos. 5 Primary ball valve (shown closed)
- Pos. 6 Pressure gauge
- Pos. 7 Relief valve
- Pos. 8 Syphon entry valve

- Close the **primary ball valve** and **low pressure relief valve** ([1/Fig. 51](#))

Dewar syphon preparation

Fig. 52 Dewar syphon

- Attach extensions as necessary, including the sludge inhibitor, to the dewar syphon leg.



- Pos. 1 Syphon flow valve
- Pos. 2 Syphon leg
- Pos. 3 Bayonet connection (male)
- Pos. 4 Extension leg
- Pos. 5 Sludge inhibitor
- Pos. 6 Syphon vacuum valve

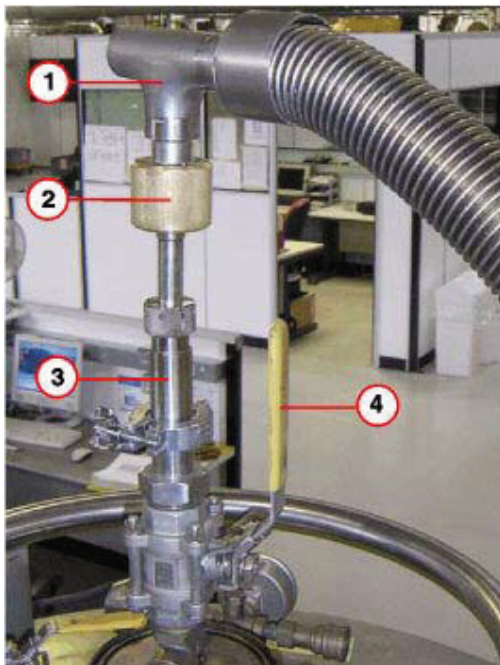
NOTE

Do not open the syphon vacuum valve (6).

Pre-cool the dewar syphon

Fig. 53 Dewar syphon fitment

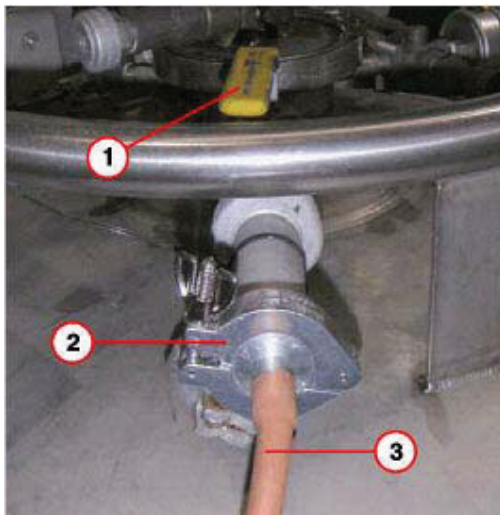
- Open the **syphon flow valve** (pos 2) two turns.
- Insert the syphon leg into the **dewar syphon adaptor**, (pos 3) and tighten the ring nut and seal in order to create a gas seal that the syphon can slide through. (The seal does not have to be completely leak-tight at this stage).
- Open the **syphon entry valve** (pos 4), and slowly slide the syphon into the dewar. Gas will begin to flow through the syphon.



- Pos. 1 Syphon
 Pos. 2 Syphon flow valve
 Pos. 3 Dewar syphon adaptor
 Pos. 4 Syphon entry valve

- Push the syphon fully down into the dewar, (very slowly to minimize pressure build-up and flash losses), until the sludge inhibitor touches the bottom.
- Fully tighten the ring nut and seal.
- Allow the syphon to cool until a dense white "cascading plume" appears. **Do not** allow high pressure to build in the dewar, only enough to cool the syphon.
- When the syphon is cold, close the **syphon flow valve** (2/Fig. 53) and move the dewar close to the magnet so that the syphon can be coupled without excessive bending.
- Fit the **helium regulator** to the helium gas bottle.

Fig. 54 Dewar gas connection



- Pos. 1 Primary ball valve
 Pos. 2 NW25 clamp and seal
 Pos. 3 Gas supply line

- Ensure that the **gas supply line** (pos 3) is long enough to allow the dewar to be moved close to the magnet, if the magnet is at field.
- Purge the helium **gas supply line** of air.
- Connect it to the **primary ball valve** (1) on the dewar, using an NW25 clamp and seal (pos 2).

- Open the **primary ball valve** (5/Fig. 51) and pressurize the dewar to **0.27bar (4psig)**.



Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.



Only non-magnetic equipment may be taken into the magnet room.

Never transport the GHe steel cylinder into the magnet room.

NOTE

If the F3 fuse breaker is turned off during the filling procedure, there will be no power to the MSUP, and therefore no real time measurements can be taken.

Verifying the electronic low pressure switch

This procedure may be carried out in conjunction with any other process involving depressurization of the helium vessel, i.e. fitting the service syphon, cold head replacement.

Fig. 55 Mk2 Pressure Control Unit



Pressure control of the helium vessel is managed by the MSUP, via the **Pressure Control Unit**.

The low pressure switch (pos 3) must be verified **whenever the magnet is depressurized**.

When pressure in the helium vessel drops to within 0.2psi above atmospheric pressure, the **electrical pressure switch** will open and turn **off** the compressor.

As the helium vessel is re-pressurized, the switch closes and the compressor will re-start.

- | | |
|--------|---|
| Pos. 1 | Electrical pressure transducer inside (absolute pressure) |
| Pos. 2 | Analogue linear gauge (relative pressure) |
| Pos. 3 | Electrical Pressure Switch (Cold Head Compressor off) |

NOTE

The pressure switch is factory calibrated to operate at 0.2psi above atmospheric pressure.

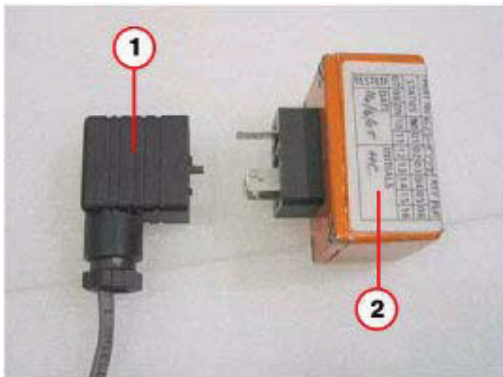
NOTE

Perform this check after filling helium when the magnet syphon is removed. The filling port is closed. Refrigerator is switched on.

- Check the **compressor** is switched on and running.
- Check **magnet pressure** is normal (15.3psia) (using the hand held display).
- De-activate the **pressure heater**:-
 - Either - Fitting the **pressure service tool**, (Fig. 56)
 - Or - Remove power from the MSUP (by removing **X3** from the MSUP).

Fig. 56 Pressure service tool electrical connection

- Fit the **pressure service tool** to the rear



Pos. 1 Pressure sensor cable

Pos. 2 Pressure service tool

pressure sensor cable.

- The Alarm Box will sound **PRESSURE ERROR**.
- Press the **ACKNOWLEDGE** button to cancel the audible alarm.
- The pressure reading will be set to **~18psia**, which will disable the pressure heater.

NOTE

Do not **remove F3** in the **ACC** - this will shut down the compressor.

De-pressurize the helium vessel:-

- Open the bypass valve.
- Depressurize until the compressor switches **off**.¹

Re-pressurize the helium vessel

- **Immediately** close the **bypass valve**.
 - **Remove** the **pressure service tool**, (or re-connect **X3** on the MSUP) to activate the pressure heater.
 - Check that the **compressor** activates automatically as the pressure rises above atmospheric.

NOTE

Check that the bypass valve is locked securely in the closed position.

NOTE

Contact the Magnet HSC if the test fails.

Magnet with Mk1 Venting System

Fit the magnet service syphon

CAUTION

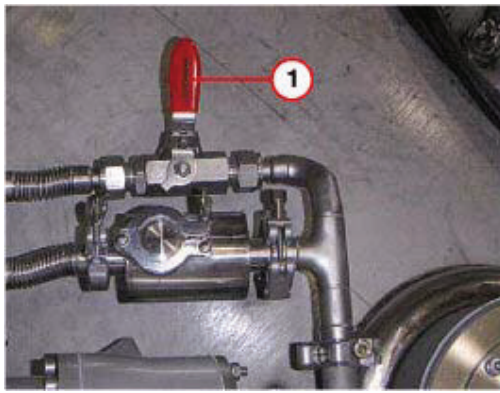
Pressurized helium vessel.

Failure to observe the following safety information may result in physical injury.

- ➡ Do not disconnect any of the fittings without first depressurizing the helium vessel.

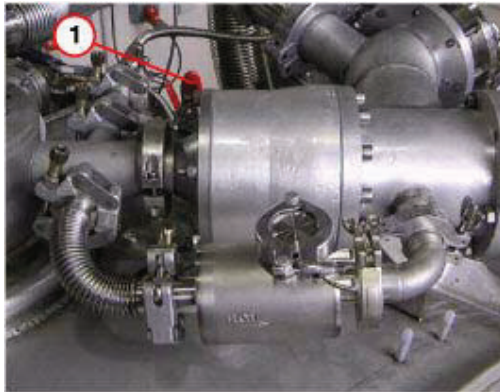
De-pressurize the helium vessel:-

- De-activate the **pressure heater**:-
 - Either - Fit the **pressure service tool**, ([Fig. 56](#))
 - Or - Remove power from the MSUP (by removing **X3** from the MSUP).
- [Fig. 57](#) OR105 Mk1 Venting bypass valve
 - **Open** the **magnet venting bypass valve** slowly to depressurize the helium vessel.
 - Do not open fully until most of the pressure has been released.
 - Depressurize until the compressor switches off.



Pos. 1 Bypass valve (shown closed)

Fig. 58 OR122 Mk1 Venting bypass valve



Pos. 1 Bypass valve (shown closed)

- Fig. 59 Mk1 Pressure manometer

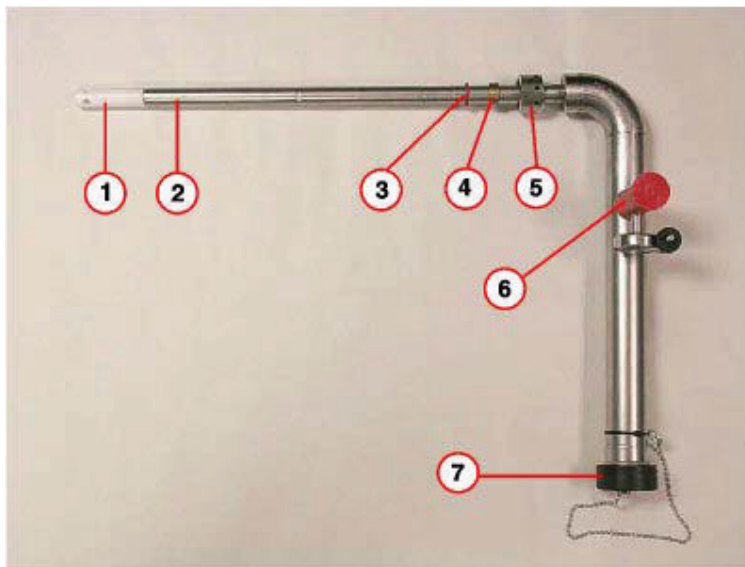


Pos. 1 Scale of helium pressure

- Check the helium vessel has a small positive pressure.

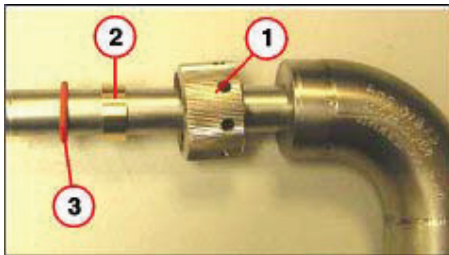
- Assemble the Service Syphon.

Fig. 60 Magnet service syphon assembly



- Pos. 1 40mm PTFE syphon diffuser
- Pos. 2 130mm syphon extension leg (2off)
- Pos. 3 Silicon 'O'-ring
- Pos. 4 Brass spacer
- Pos. 5 Ring nut
- Pos. 6 Vacuum valve
- Pos. 7 Syphon end cap

Fig. 61 Assembling the Ring Nut Seal



- Pos. 1 Ring nut
- Pos. 2 Brass spacer
- Pos. 3 Silicon O-ring

- Fit the ring nut, brass spacer and the 'O'-ring to the **service syphon leg**.

- Fit the 130mm long **extension legs** (2 off) ([2/Fig. 60](#)), and the **40mm PTFE diffuser**. ([1/Fig. 60](#))

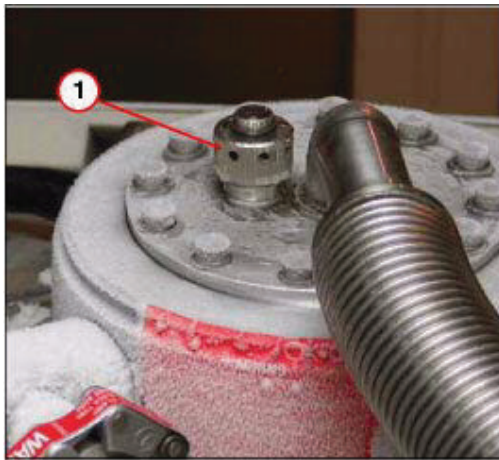
NOTE

There is a risk of quenching the magnet if the 40mm PTFE diffuser is not fitted.

- Remove the **syphon end cap** ([7/Fig. 60](#)).
- **Check the ceiling height**. If necessary fit the syphon leg support clamp - refer to ([Low Ceiling Height](#))

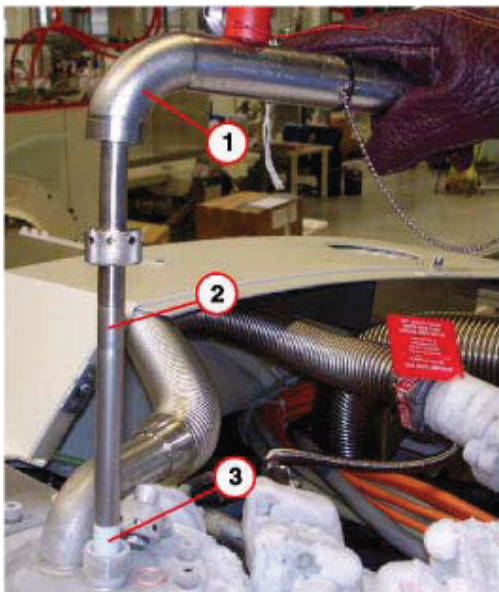
Fig. 62 Mk1 Syphon entry port

- **Remove** the **syphon blanking plug** using the syphon 'C' spanner. ([9/Fig. 49](#))
- Warm the blanking plug using a hot air gun to ease removal.



Pos. 1 Syphon blanking plug

Fig. 63 Helium syphon



Pos. 1 Service Syphon
Pos. 2 Two Extension Legs
Pos. 3 PTFE Syphon Diffuser

- **Slowly** lower the service syphon fully into the neck.
- Tighten the **syphon port ring nut** ([1/ Fig. 61](#)) by hand, then apply a half turn using the syphon 'C' spanner to form an air-tight seal.

NOTE There is a risk of quenching the magnet if the syphon is inserted too quickly.

- Re-fit the **syphon end cap** ([7/ Fig. 60](#)).
- **Close** the bypass valve.

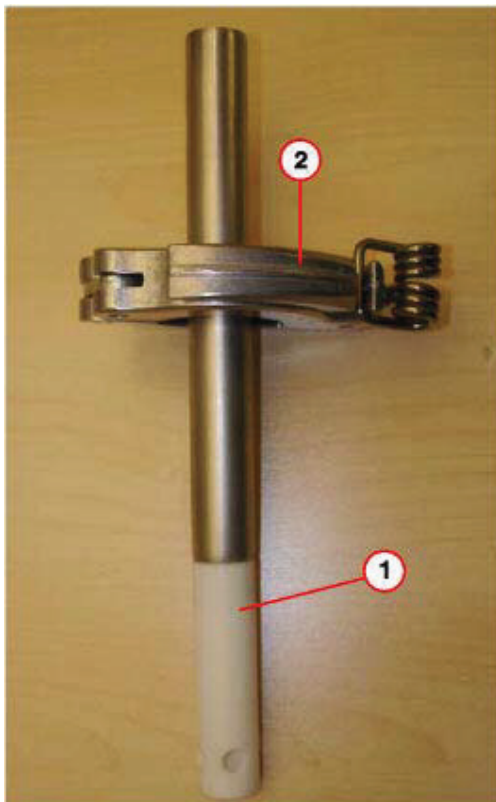
Low Ceiling Height

Where magnets have been installed into rooms with a low ceiling height, it can be difficult to fit the service syphon. To avoid dropping the extension leg(s) down into the syphon port, the **syphon leg support clamp** should be used ([8/ Fig. 49](#)).

Fig. 64 Syphon configuration for low ceiling height

Refer to ([Fig. 49](#))

- Fit the syphon leg support clamp onto the PTFE syphon diffuser leg and lower it into the magnet syphon entry port.



- Pos. 1 PTFE Syphon Diffuser
Pos. 2 Syphon leg support clamp

- Screw on the second extension leg.
- Move the syphon clamp up the extension leg to allow room for the syphon body.
- Ensure the ring nut, brass spacer and 'O'-ring are fitted correctly, then screw the diffuser leg to the syphon body.
- Remove the syphon clamp and lower the syphon fully into the syphon entry.
- Screw the syphon port nut into place.

NOTE Use the clamp for removing the syphon.

Pre-cool the service syphon

NOTE The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use. (Refer to [Dewar preparation](#))

Cool down the **service syphon** as follows:-

- Ensure the **syphon end cap** is fitted ([7/Fig. 60](#)), and the bypass valve is **closed**.
- Activate the pressure heater by removing the **pressure service tool** ([Fig. 56](#)) to achieve a positive pressure of about **0.3psia** in the magnet.
- **Re-connect** the pressure service tool.

Fig. 65 Syphon venting (cascading plume)



- Remove the service **syphon end cap** and allow the syphon to vent until it is fully cold and a dense white "cascading plume" appears.

- **CAUTION** Risk of cold burns.
A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- ➡ Stand clear of the syphon joint when disconnecting.
- ➡ Wear protective gloves and safety glasses.

Liquid helium transfer

- Check the gas supply pressure is set at **0.27bar (4psig)**.
- As the **service syphon** nears the point of being fully cold, open the **dewar syphon flow valve** half way (2 turns) (Fig. 67), and pre-cool the syphon again until a dense white "cascading plume" appears

Fig. 66 Dewar and service syphon coupling



- Pos. 1 Service syphon
 Pos. 2 Lock nut
 Pos. 3 Dewar syphon vacuum valve
 Pos. 4 Dewar syphon

- When **both** syphons are giving a dense white "cascading plume" of helium gas, couple them together and secure the lock nut.

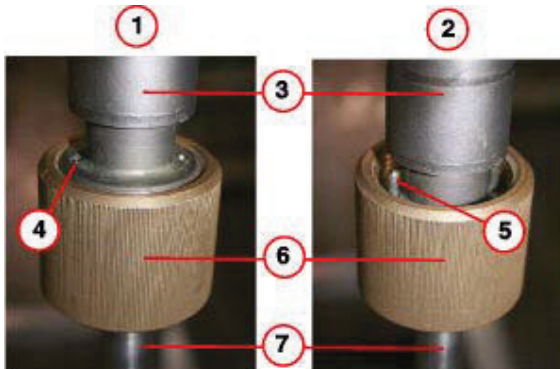


Risk of quench

Warm helium gas can cause a quench.

- ➡ Do not leave the transfer unattended.
- ➡ Replace syphons that show signs of poor vacuum.

Fig. 67 Dewar syphon flow valves - open and closed positions Fig. 68 Helium filling



- Pos. 1 FULLY OPEN position
 Pos. 2 FULLY CLOSED position
 Pos. 3 Stainless steel body



- Pos. 4 Guide pin flush with body
- Pos. 5 Guide pin
- Pos. 6 Brass nut
- Pos. 7 Syphon leg

- Maximize the transfer by fully opening the **dewar syphon flow valve**.
 - ➔ The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (4/ Fig. 67)
 - ➔ From fully closed to fully open, is **4 turns**.
- Re-fit the **pressure service tool** (Fig. 56) to de-activate the pressure heater.

Fig. 69 MSUP Handheld Display Unit



- Pos. 1 LCD Display
- Pos. 2 Sample button
- Pos. 3 Display scroll button
- Pos. 4 25 way 'D' connector

- Take an initial helium level measurement from the **MSUP display unit**.
- Repeat the level measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - If not, stop the transfer and investigate the reason for poor transfer efficiency.

NOTE Opening the bypass valve may help to reduce the back pressure, enabling more efficient helium transfer.

NOTE Maintain a pressure of about 0.27bar (4psig) at the helium dewar.
Too much or too little pressure may result in inefficiency.

At **4psig**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 litre dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**

CAUTION Risk of quench
Warm helium gas can cause a quench.

- ➔ Do not leave the transfer unattended.
- ➔ Replace syphons that show signs of poor vacuum.

Stopping the Helium Transfer

Fig. 70 Dewar manometer

- As soon as the level meter reads 95%, or the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon.
Do not fill above 95%.



- Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - **if left open there is a risk of quenching.** ([2/Fig. 53](#))
- Set the dewar gas pressure regulator to **zero**.
- Start de-pressurization of the dewar.
Refer to ([Changing supply dewars during transfer](#))

NOTE

Do not fill above 95%.

The magnet cannot be ramped at a higher helium level.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.



Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon.

Removing the Service Syphon

- **Open** the **venting bypass valve** very slowly to depressurize the helium vessel to **0.1 psia (7 mbar)** above atmospheric pressure.
- Warm the syphon joint to room temperature (using a heat gun), to remove ice.
- **Close** the **venting bypass valve** before disconnecting the syphons.
- **Disconnect** the syphons.

**CAUTION**

Risk of cold burns.

Cold gas will exhaust from the syphons when they are disconnected.



Stand clear of the syphon joint when disconnecting.

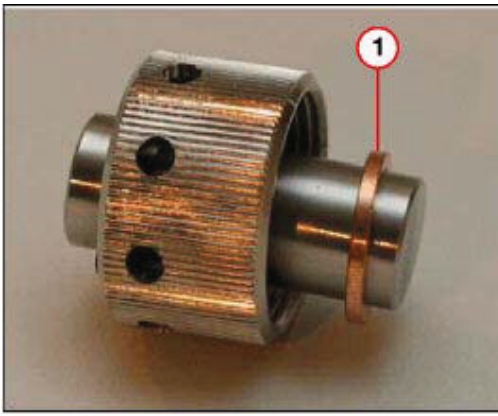


Wear protective gloves.

- Close off the end of the service syphon with the **syphon end cap** ([7/Fig. 60](#)).
- Remove the service syphon from the magnet **immediately** after de-coupling the syphons.

Fig. 71 Mk1 syphon blanking plug

Fig. 72 Mk1 Syphon entry port



Pos. 1 Copper seal



Pos. 1 Syphon 'C' spanner

- Fit the **syphon blanking plug** using a new copper seal.
- Warm up the threads on the plug and **carefully** tighten with the syphon 'C' spanner.
 - There may be ice on the threads - **do not use undue force**. Damaged threads will result in a leak.
- Warm the **top plate** and **syphon plug** using a heat gun. Re-tighten the plug.



CAUTION Danger of air ingress.

Internal ice blockage could cause excessive pressure.



Warm up syphon port plug sufficiently to ensure there is no ice formed on the threads.

- **Remove** the **pressure service tool** ([Fig. 56](#)) and **re-connect** the **pressure sensor cable** to the pressure switch.
- Check that all **pressure alarms** are de-activated and that the **pressure heater** is operational.
- Switch on the **compressor** and allow pressure in the magnet to rise.
- Check that the **pressure heater** activates with a 20% duty cycle. The yellow LED on the MSUP should also illuminate at the same duty cycle.
- Check that the **compressor** activates automatically as the pressure rises above atmospheric.
- Warm the turret and venting valves to room temperature with a hot air gun.

NOTE

Wait at least 2 hours after completion of a helium fill before attempting to ramp the magnet.

- Complete depressurizing the dewar by opening the **primary ball valve**. ([5/Fig. 51](#))
- When at the gas pressure is at **zero**, remove the dewar syphon.
- **Immediately** close the **syphon entry port** on top of the dewar. ([4/Fig. 51](#))
- **Close** the **primary ball valve** ([5/Fig. 51](#)), and **open** the **low-pressure relief valve**. ([1/Fig. 51](#))
- Warm both syphons with the heat gun until dry, and store safely.

Magnet with Mk2 Venting System

Fit the magnet service syphon



CAUTION Pressurized helium vessel

Failure to observe the following safety information may result in physical injury.



Do not disconnect any of the fittings without first depressurizing the helium vessel.

De-pressurize the helium vessel:-

- De-activate the **pressure heater**:-
 - Either - Fit the **pressure service tool** ([Fig. 56](#))
 - Or - Remove power from the MSUP (by removing **X3** from the MSUP).

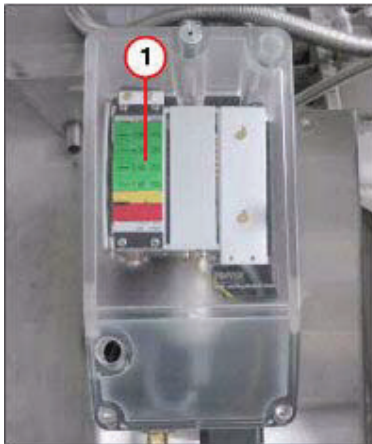
Fig. 73 Mk2 Venting bypass valve



Pos. 1 Bypass valve (show open)

- **Open the magnet venting bypass valve** slowly to depressurize the helium vessel.
- Do not open fully until most of the pressure has been released.
- Depressurize until the compressor switches off.

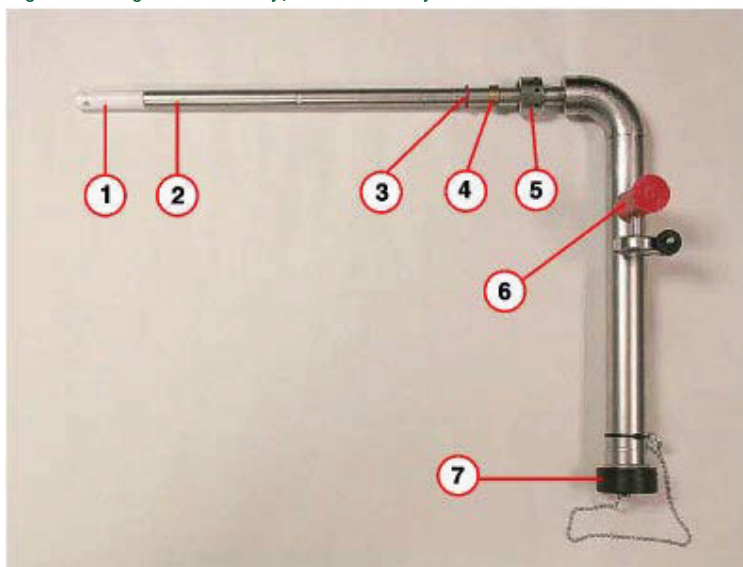
Fig. 74 Mk2 Pressure Manometer



- Check the helium vessel has a small positive pressure.

- Assemble the Service Syphon.

Fig. 75 Magnet service syphon assembly

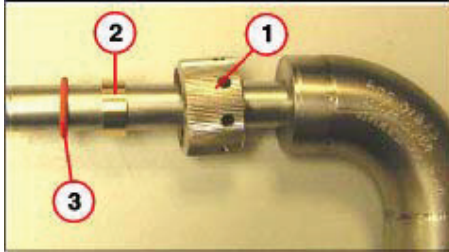


Pos. 1 40mm PTFE syphon diffuser

Pos. 2 130mm syphon extension leg (2off)

- Pos. 3 Silicon 'O'-ring
- Pos. 4 Brass spacer
- Pos. 5 Ring nut
- Pos. 6 Vacuum valve
- Pos. 7 Syphon end cap

Fig. 76 Assembling the Ring Nut Seal



- Pos. 1 Ring nut
- Pos. 2 Brass spacer
- Pos. 3 Silicon O-ring

- Fit the ring nut, brass spacer and the 'O'-ring to the **service syphon leg**.

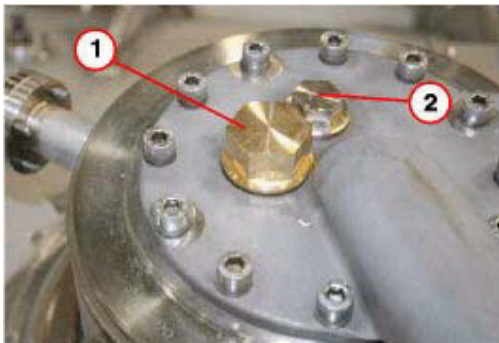
- Fit the 130mm long **extension legs** (2 off) and the **40mm PTFE diffuser**. (2/Fig. 75)

NOTE

There is a risk of quenching the magnet if the 40mm PTFE diffuser is not fitted.

- Remove the **syphon end cap** (7/Fig. 75).
- **Check the ceiling height**. If necessary fit the syphon leg support clamp - refer to ([Low Ceiling Height](#))

Fig. 77 Mk2 Syphon port entry



- Pos. 1 Syphon blanking plug
- Pos. 2 Sight glass

- **Remove** the **syphon blanking plug** using the syphon 'C' spanner (9/Fig. 49).
- On later systems the blanking plug has **hexagonal** flats and a non-magnetic 30mm spanner is needed.

Fig. 78 Mk2 Venting system



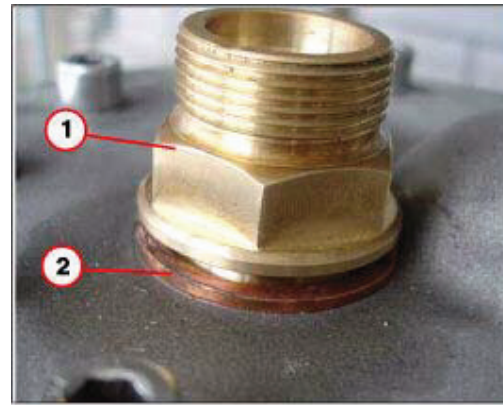
- Warm the blanking plug using a hot air gun to ease removal.

Fig. 79 Mk2 Syphon entry port

Fig. 80 Mk2 Fill adaptor



Pos. 1 Syphon blanking plug



Pos. 1 Fill adaptor

Pos. 2 Copper seal

Fig. 81 Mk2 Fill adaptor fitment



Fig. 82 Mk2 Syphon fitment



NOTE

There is a risk of quenching the magnet if the syphon is inserted too quickly.

Fig. 83 Mk2 Syphon fitment

- Fit the **brass syphon fill adaptor** and a new copper seal.

- Guide the syphon into the **syphon fill adaptor**.

- **Slowly** lower the service syphon fully into

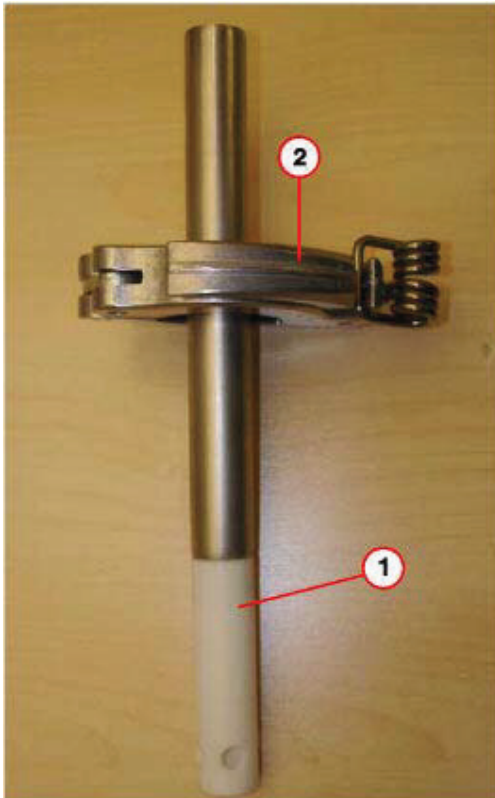


- the neck .
- Tighten the **syphon port ring nut** by hand, then apply a half turn using the syphon 'C' spanner to form an air-tight seal.
- Re-fit the **syphon end cap**. ([7/Fig. 75](#))
- **Close** the bypass valve.

Low Ceiling Height

Where magnets have been installed into rooms with a low ceiling height, it can be difficult to fit the service syphon. To avoid dropping the extension leg(s) down into the syphon port, the **syphon leg support clamp** should be used.

Fig. 84 Syphon configuration for low ceiling height



- Pos. 1 PTFE Syphon Diffuser
Pos. 2 Syphon leg support clamp

Refer to ([Fig. 49](#))

- Fit the syphon leg support clamp onto the PTFE syphon diffuser leg and lower it into the magnet syphon entry port.
- Screw on the second extension leg.
- Move the syphon clamp up the extension leg to allow room for the syphon body.
- Ensure the ring nut, brass spacer and 'O'-ring are fitted correctly, then screw the diffuser leg to the syphon body.
- Remove the syphon clamp and lower the syphon fully into the syphon entry.
- Screw the syphon port nut into place.

NOTE

Use the clamp for removing the syphon.

Pre-cool the service syphon

NOTE

The helium dewar must be positioned close to the magnet with the dewar syphon fitted, ready for use. (Refer to ([Dewar preparation](#)))

Cool down the **service syphon** as follows:-

- Ensure the **syphon end cap** ([7/7Fig. 75](#)) is fitted and the bypass valve is **closed**.
- Activate the pressure heater by removing the **pressure service tool** ([Fig. 56](#)) to achieve a positive pressure of about **0.3psia** in the magnet.
- **Re-connect** the pressure service tool.

Fig. 85 Syphon venting (cascading plume)



- Remove the service **syphon end cap** and allow the syphon to vent until it is fully cold and a dense white "cascading plume" appears.

CAUTION

Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- ➔ Stand clear of the syphon joint when disconnecting.
- ➔ Wear protective gloves and safety glasses.

CAUTION

Risk of quench

Warm helium gas can cause a quench.

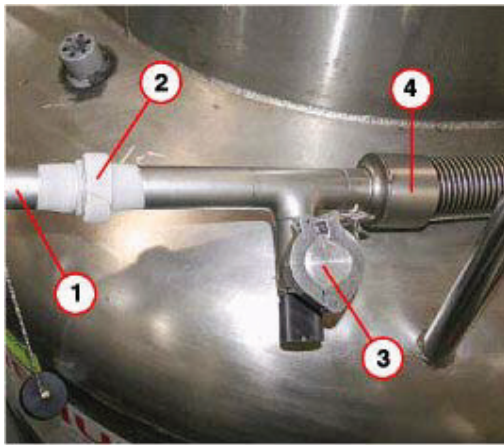
- ➔ Do not leave the transfer unattended.
- ➔ Replace syphons that show signs of poor vacuum.

Liquid helium transfer

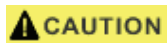
- Check the gas supply pressure is set at **0.27bar (4psig)**.
- As the **service syphon** nears the point of being fully cold, open the **dewar syphon flow valve** half way (2 turns) ([Fig. 87](#)), and pre-cool the syphon again until a dense white "cascading plume" appears

Fig. 86 Dewar and service syphon coupling

- When **both** syphons are giving a dense white "cascading plume" of helium gas, couple them together and secure the lock nut.



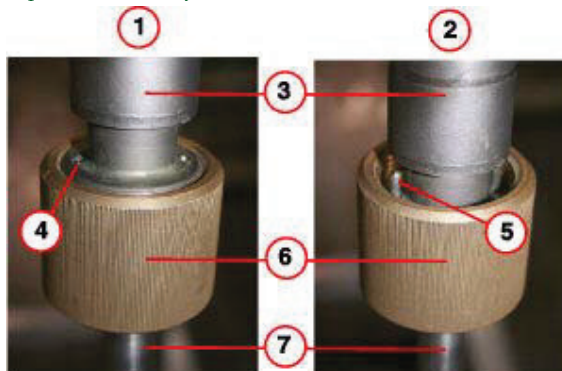
- Pos. 1 Service syphon
 Pos. 2 Lock nut
 Pos. 3 Dewar syphon vacuum valve
 Pos. 4 Dewar syphon



Risk of quench if magnet is at field.
 Warm helium gas will cause a quench.

➔ Take care that both syphons are fully cold.

Fig. 87 Dewar syphon flow valves - open and closed positions Fig. 88 Helium filling



- Pos. 1 FULLY OPEN position
 Pos. 2 FULLY CLOSED position
 Pos. 3 Stainless steel body
 Pos. 4 Guide pin flush with body
 Pos. 5 Guide pin
 Pos. 6 Brass nut
 Pos. 7 Syphon leg



- Maximize the transfer by fully opening the **dewar syphon flow valve**.
 - ➔ The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (4/Fig. 87)
 - ➔ From fully closed to fully open, is **4 turns**.
- Re-fit the **pressure service tool** to de-activate the pressure heater (Fig. 56).

Fig. 89 MSUP Handheld Display Unit



- Pos. 1 LCD Display
 Pos. 2 Sample button
 Pos. 3 Display scroll button
 Pos. 4 25 way 'D' connector

- Take an initial helium level measurement from the **MSUP display unit**.
- Repeat the level measurement every 5 minutes. An increase in level should be seen within 20 minutes.
 - If not, stop the transfer and investigate the reason for poor transfer efficiency.

NOTE

Opening the bypass valve may help to reduce the back pressure, enabling more efficient helium transfer.

NOTE

Maintain a pressure of about 0.27bar (4psig) at the helium dewar.
 Too much or too little pressure may result in inefficiency.

At **4psig**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 litre dewar will empty in 25-30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately, the syphon may need to be replaced.**

CAUTION

Risk of quench

Warm helium gas can cause a quench.

- ➔ Do not leave the transfer unattended.
- ➔ Replace syphons that show signs of poor vacuum.

Stopping the Helium Transfer

Fig. 90 Dewar manometer



- As soon as the level meter reads 95%, or the dewar is empty, stop the transfer by closing the flow valve at the dewar syphon. **Do not fill above 95%.**
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties - if left open there is a risk of quenching. (2/Fig. 92)
- Set the dewar gas pressure regulator to **zero**.
- Start de-pressurization of the dewar. Refer to ([Changing supply dewars during transfer](#))

NOTE

Do not fill above 95%.

The magnet cannot be ramped at a higher helium level.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.

- ➔ Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon.

Removing the Service Syphon

- **Open** the **venting bypass valve** ([Fig. 73](#)) very slowly to depressurize the helium vessel to **0.1 psia (7 mbar)** above atmospheric pressure .
- Warm the syphon joint to room temperature (using a heat gun), to remove ice.
- **Close** the **venting bypass valve** before disconnecting the syphons.
- **Disconnect** the syphons.

CAUTION

Risk of cold burns.

Cold gas will exhaust from the syphons when they are disconnected.

- ➔ Stand clear of the syphon joint when disconnecting.
- ➔ Wear protective gloves.

- Close off the end of the service syphon with the **syphon end cap** ([7/ Fig. 75](#)).
- Remove the service syphon from the magnet **immediately** after de-coupling the syphons.

Fig. 91 Mk2 Blanking plug fitment



- Remove the brass **syphon fill adaptor** from the turret top plate and fit the **syphon blanking plug** using a new copper seal.
- Warm up the threads on the plug and **carefully** tighten them with the 30mm non-magnetic spanner.
 - There may be ice on the threads-**do not use undue force**. Damaged threads will result in a leak.

- Warm the **top plate** and **syphon plug** using a heat gun. Re-tighten the plug.

CAUTION

Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- ➔ Warm up syphon port plug sufficiently to ensure there is no ice formed on the threads.

- **Remove** the **pressure service tool** ([Fig. 56](#)) and **re-connect** the **pressure sensor cable** to the pressure switch.
- Check that all **pressure alarms** are de-activated and that the **pressure heater** is operational.
- Switch on the **compressor** and allow pressure in the magnet to rise.
- Check that the **pressure heater** activates with a 20% duty cycle. The yellow LED on the MSUP should also illuminate at the same duty cycle.
- Check that the **compressor** activates automatically as the pressure rises above atmospheric.
- Warm the turret and venting valves to room temperature with a hot air gun.

NOTE

Wait at least 2 hours after completion of a helium fill before attempting to ramp the magnet.

- Complete depressurizing the dewar by opening the **primary ball valve**. (5/Fig. 93)
- When at the gas pressure is at **zero**, remove the dewar syphon.
- **Immediately** close the **syphon entry port** on top of the dewar. (4/Fig. 93)
- **Close** the **primary ball valve** (5/Fig. 93), and **open** the **low-pressure relief valve**. (1/Fig. 93)
- Warm both syphons with the heat gun until dry, and store safely.

Changing supply dewars during transfer

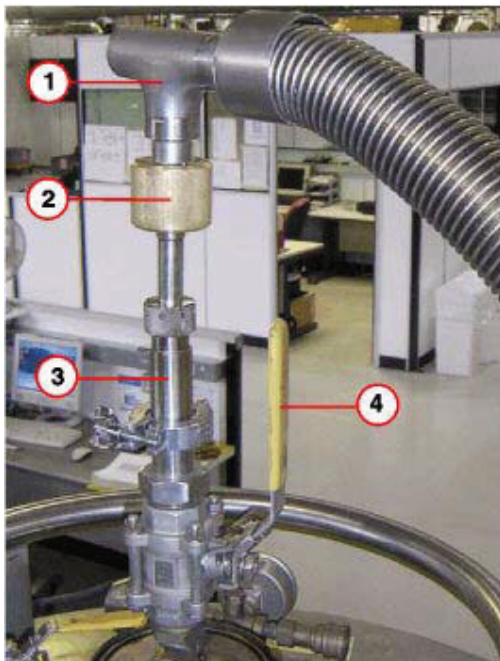
NOTE

Whistling of the syphon (a high pitched sound), or a noticeable drop in dewar pressure are indications of an empty dewar.

If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one.

Ensure that a full dewar is available and depressurized, ready to receive the syphon.

Fig. 92 Dewar syphon fitment



- When the dewar empties, stop the transfer **immediately** by closing the **dewar syphon flow valve** (pos 2).

- | | |
|--------|----------------------|
| Pos. 1 | Syphon |
| Pos. 2 | Syphon flow valve |
| Pos. 3 | Dewar syphon adaptor |
| Pos. 4 | Syphon entry valve |

CAUTION

Risk of quench

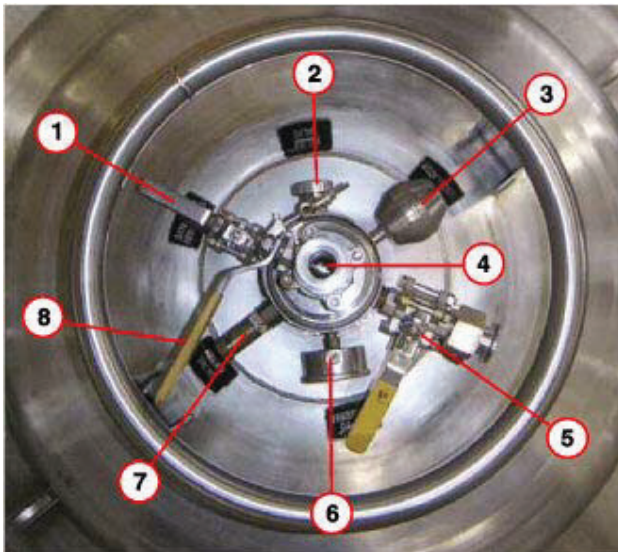
Failure to close the valve immediately, or leaving the valve open when magnet is at field, may result in a quench.



Close the valve immediately.

- Turn off the gas supply and vent the storage dewar down to atmospheric pressure. **This must be done in a well ventilated area - NOT in the imaging suite.**

Fig. 93 Dewar Ports (typical)



- Pos. 1 Low pressure relief valve (shown open)
- Pos. 2 Relief valve
- Pos. 3 Anti-oscillator
- Pos. 4 Syphon entry port (shown closed)
- Pos. 5 Primary ball valve (shown closed)
- Pos. 6 Pressure gauge
- Pos. 7 Relief valve
- Pos. 8 Syphon entry valve

- **Close** the **primary ball valve**, ([5/Fig. 93](#)), and **open** the **low pressure relief valve**. ([1/Fig. 93](#))
- Using a heat gun, warm the syphon joint to room temperature.
- **Disconnect** the dewar from the service syphon.



Risk of cold burns.

A "cascading plume" of cold gas will exhaust from the syphons when they are disconnected.

- ➡ Stand clear of the syphon joint when disconnecting.
- ➡ Wear protective gloves and safety glasses.

- Remove any moisture from the end of the **service syphon** with paper towelling.
- Fit the **service syphon end cap** ([7/Fig. 75](#)).



Danger of air ingress.

Internal ice blockage could cause excessive pressure.

- ➡ Ensure the service syphon end cap is fitted immediately and securely.

- Open and close the **primary ball valve** ([5/Fig. 93](#)), to ensure the empty dewar is depressurized.
- When the gas pressure is at **zero**, remove the dewar syphon from the empty dewar and **immediately** insert the syphon leg into the full dewar.
- **Close** the **syphon entry port** ([4/Fig. 93](#)), and **open** the **low pressure relief valve** ([1/Fig. 93](#)).
- Pre-cool the dewar syphon ([Pre-cool the dewar syphon](#)).

Final work steps

After transfer, check that:-

- The **magnet bypass valve** is **closed** and **locked**.
- The **pressure heater(s)** are operating correctly.
- The pressure heater LED on the MSUP is illuminated.

- No alarm signals are evident on the hand held MSUP display unit, or the Alarm Box.
- The **compressor** is switched **on** and running.
- The **magnet pressure** is stable at **15.3 psia**.
- Warm the **service syphon** with the heat gun.
- Warm the turret and venting system with heat gun.
- **Disconnect** the **MSUP display unit** from the MSUP.
- Check the filling port and venting system for leaks. (Refer to **Turret Leak Check procedure**).
- Check around the 4K cold head flange for leaks.

Bottom filling the magnet

Bottom filling of the magnet will be required if:-

- the liquid helium level falls to **zero** percent.
- the temperatures measured from the carbon resistors and coil diodes, indicates temperatures **no warmer than** liquid nitrogen temperature. Refer to Table 1 ([Tab. 3 Expected Carbon Resistor Values](#))

The magnet must be filled through the **magnet syphon cone** to allow filling from the bottom of the helium vessel.

NOTE

Only CSEs trained by Siemens in filling the magnets are allowed to perform this work.

NOTE

Inform Magnet HSC immediately if the helium level is zero. Magnet coil temperatures must be confirmed before continuing with the helium fill.

Possible causes for this include:-

- Post magnet quench, ramping or from persistence.
- The magnet has warmed up.
- Faulty helium level probes.
- Alarm box buttons still depressed.
- Pressure heaters left on.
- Long transportation and storage times.

Checking Magnet Temperatures

To recover a magnet with a **bottom helium fill**, the following temperatures must be checked:-

- Use the **MSUP hand held display unit** to measure:-
 - **Helium level** values
 - **Carbon resistor** values

Tab. 3 Expected Carbon Resistor Values

41 way plug Measured in Ohms	Expected Values Room Temperature	Expected Values 77K Temperature	Expected Values 4.2K Temperature
Carbon One a - e	140 - 180	170 - 200	700 - 1500
Carbon Two b - e	120 - 160	130 - 180	700 - 1500
Carbon Three c - e	120 - 160	130 - 180	700 - 1500
Carbon Four* d - e	120 - 160	130 - 180	700 - 1500

NOTE

*Carbon 4 is for measuring helium gas temperatures during ramping and is not applicable to filling.

Bottom fill procedure

De-pressurize the helium vessel:-

- De-activate the **pressure heater**:-
 - Either - Fit the **pressure service tool** ([Fig. 56](#))

- Or - Remove power from the MSUP (by removing **X3** from the MSUP).
- Open the **magnet venting bypass valve** slowly to depressurize the helium vessel.
 - ➡ Do not open fully until most of the pressure has been released.
- Depressurize until the compressor switches off.
- Check the helium vessel has a small positive pressure.
- **Remove the syphon blanking plug.** Warm the plug, using a hot air gun to ease removal.



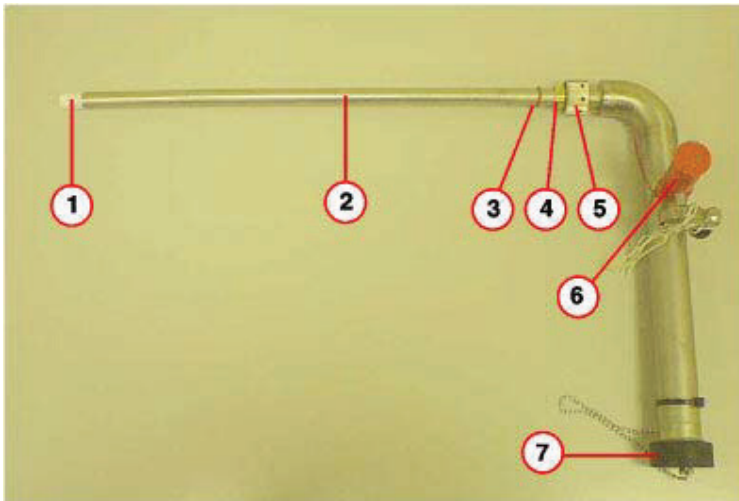
Risk of cold burns.

Cold gas will exhaust from the syphons when they are disconnected.

- ➡ Stand clear of the syphon joint when disconnecting.
- ➡ Wear protective gloves.

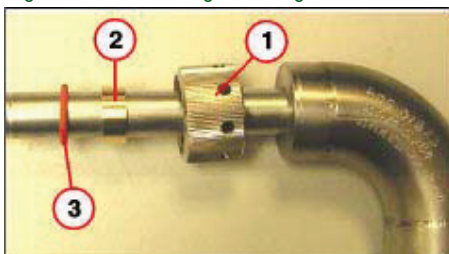
- **Mk2 only** - Fit the brass syphon fill adaptor and a new copper seal. ([Fig. 80](#))

Fig. 94 Magnet service syphon assembly - bottom filling



- | | |
|--------|-------------------------------------|
| Pos. 1 | PTFE seal (bottom fill) |
| Pos. 2 | 130mm syphon extension Legs (3 off) |
| Pos. 3 | Silicon 'O'-ring |
| Pos. 4 | Brass spacer |
| Pos. 5 | Ring nut |
| Pos. 6 | Vacuum valve |
| Pos. 7 | Syphon end cap |

Fig. 95 Assembling the Ring Nut Seal



- | | |
|--------|----------------|
| Pos. 1 | Ring nut |
| Pos. 2 | Brass spacer |
| Pos. 3 | Silicon O-ring |

- Fit the ring nut, brass spacer and the 'O'-ring to the **service syphon leg**.

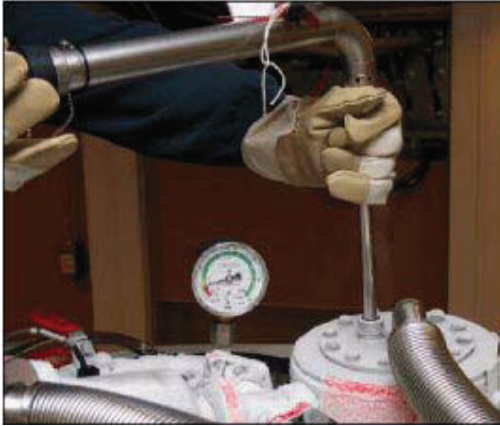
- Fit the 130mm long **extension legs** (3 off) ([2/ Fig. 94](#)), and the **PTFE seal (bottom fill)**. ([1/ Fig. 94](#))

NOTE

The minimum length required to reach the syphon cone is 485mm (+/- 5mm)

- Remove the **syphon end cap** ([7/ Fig. 94](#))
- **Check the ceiling height.** If necessary fit the syphon leg support clamp - refer to ([Low Ceiling Height](#))

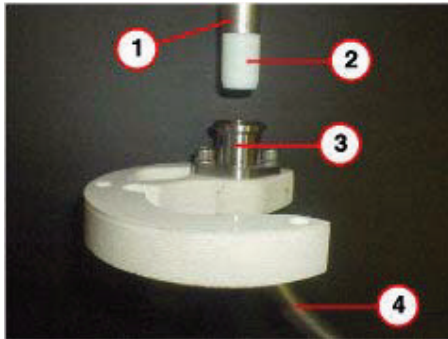
Fig. 96 Syphon



- Slowly lower the service syphon fully into the neck.

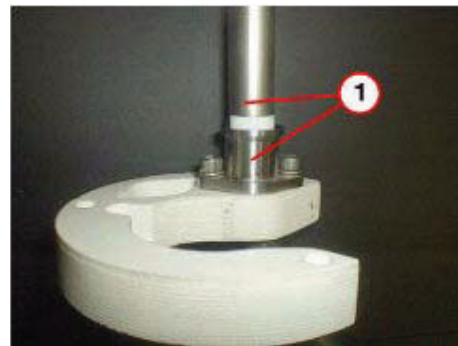
- The syphon leg must locate fully into the **syphon cone** - make sure it is **fully engaged**.

Fig. 97 Bottom fill syphon cone (pictorial view)



- Pos. 1 Syphon leg
Pos. 2 PTFE Seal (Bottom Fill)
Pos. 3 Syphon cone
Pos. 4 Syphon tube

Fig. 98 Bottom fill - syphon leg fully engaged (pictorial view)



- Pos. 1 Syphon leg - fully engaged

NOTE

Helium will be wasted if the syphon leg is not fully engaged in the syphon cone.

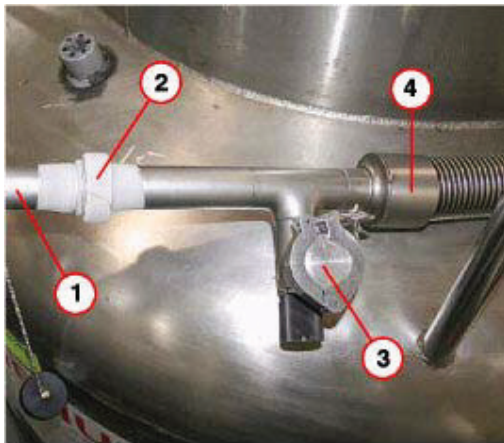
- Tighten the **syphon port ring nut** ([Fig. 95](#)) by hand, then apply a half turn, (using the appropriate spanner), to form an air-tight seal.
- Re-fit the **syphon end cap**.
- **Close** the bypass valve.
- **Pre-cool the service syphon.** ([Pre-cool the service syphon](#))

Liquid helium transfer

- Check the gas supply pressure is set at **0.27bar (4psig)**.
- As the **service syphon** nears the point of being fully cold, open the **dewar syphon flow valve** half way (2 turns) ([Fig. 100](#)), and pre-cool the syphon again until a dense white "cascading plume" appears

Fig. 99 Dewar and service syphon coupling

- When **both** syphons are giving a dense white "cascading plume" of helium gas, couple them together and secure the lock nut.



- Pos. 1 Service syphon
 Pos. 2 Lock nut
 Pos. 3 Dewar syphon vacuum valve
 Pos. 4 Dewar syphon

Fig. 100 Dewar syphon flow valves - open and closed positions

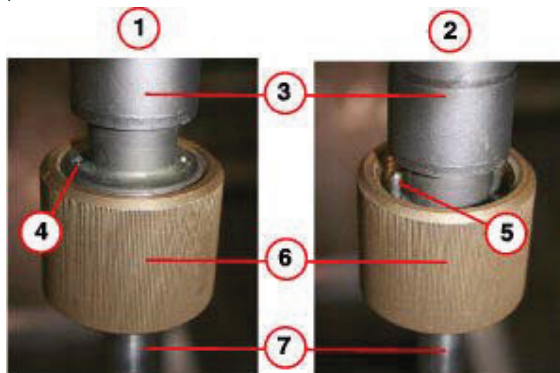


Fig. 101 Helium filling

- Pos. 1 FULLY OPEN position
 Pos. 2 FULLY CLOSED position
 Pos. 3 Stainless steel body
 Pos. 4 Guide pin flush with body
 Pos. 5 Guide pin
 Pos. 6 Brass nut
 Pos. 7 Syphon leg

- Maximize the transfer by fully opening the **dewar syphon flow valve**.
 - ➡ The flow valve is **fully open** when the two **guide pins** at the top of the brass nut are flush with the top of the stainless steel body. (4/Fig. 100)
 - ➡ From fully closed to fully open, is **4 turns**.
- **Re-fit the pressure service tool** (Fig. 56) to de-activate the pressure heater.

Fig. 102 MSUP Handheld Display Unit

- Take an initial helium level measurement from the **MSUP hand held display unit**:-
- Monitor the **carbon resistors** every 15 minutes (or as advised by Magnet HSC), to measure the temperature drop.
 - ➡ Refer to the **acceptance**



- Pos. 1 LCD Display
 Pos. 2 Sample button
 Pos. 3 Display scroll button
 Pos. 4 25 way 'D' connector

documentation (Installation Protocol) for required values.

➡ If the values do not rise, **contact Magnet HSC.**

- When the required value is reached, monitor the **helium level** using the hand held display unit.

NOTE Approximately 500-700 liters of helium is required to achieve a 10% meter reading, from nitrogen temperature.

NOTE Opening the bypass valve may help to reduce the back pressure, enabling more efficient helium transfer.

NOTE Maintain a pressure of about 0.27bar (4psig) at the helium dewar.
 Too much or too little pressure may result in inefficiency.

At **4psig**, the transfer rate is approximately 10 liters per minute. **Do not leave the transfer unattended.** A 250 litre dewar will empty in 25 - 30 minutes.

If frost or condensation forms on the body of either syphon, it is an indication of poor vacuum. **Stop the transfer immediately - the syphon may need to be replaced.**

Stopping the Helium Transfer

Fig. 103 Dewar manometer



- When the helium level meter reads 95%, allow the dewar to empty, then stop the transfer by closing the **flow valve** at the dewar syphon.
 - Whistling (high pitched sound) of the syphon or noticeable drop in dewar pressure are indications of an empty dewar.
- The **dewar syphon flow valve** must be closed **immediately** the dewar empties.
- Set the dewar gas pressure regulator to **zero**.
- Start de-pressurization of the dewar.
 Refer to ([Changing supply dewars during transfer](#))

NOTE Do not fill above 95%.
 The magnet cannot be ramped at a higher helium level.

NOTICE

Risk of quench.

Warm helium gas will cause a quench.



Stop the transfer immediately in case you hear whistling (a high pitched sound) from the syphon.

Removing the Service Syphon

- **Open** the **venting bypass valve** ([Fig. 57](#)), ([Fig. 58](#)), ([Fig. 73](#)), very slowly to depressurize the helium vessel to **0.1 psia (7 mbar)** above atmospheric pressure.
- Warm the syphon joint to room temperature (using a heat gun), to remove ice.
- **Close** the **venting bypass valve** before disconnecting the syphons.
- **Disconnect** the syphons.

**CAUTION**

Risk of cold burns.

Cold gas will exhaust from the syphons when they are disconnected.



Stand clear of the syphon joint when disconnecting.



Wear protective gloves.

- Close off the end of the service syphon with the **syphon end cap** ([7/ Fig. 75](#)).
- Remove the service syphon from the magnet **immediately** after de-coupling the syphons.
- For **Mk1** venting - first fit the **syphon blanking plug**. ([Fig. 71](#))
- For **Mk2** venting - first remove the **brass syphon fill adaptor** and then fit the **syphon blanking plug**. ([Fig. 91](#))
- Warm the extension legs using a heat gun.
- Remove the **bottom fill extension leg** and **PTFE seal**.

The magnet filling can now be completed following the normal filling procedures - see ([Magnet with Mk1 Venting System](#)) and ([Magnet with Mk2 Venting System](#)).

After a bottom fill, it is recommended that the first magnet energization is performed the next day. This will allow the magnet coils to thermally stabilize.

¹ If the verification is being carried out with another process (i.e. fitting the Magnet Service Syphon/replacing the Cold head), follow the **re-pressurization** part of this procedure when those processes are complete.